

SMOKY LAKE AGDM, AB, Canada

WMO: 712740

Lat: 54.265N

Lon: 112.504W

Elev: 680

StdP: 93.42

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.1	-28.8	-35.6	0.1	-31.8	-32.1	0.2	-28.5	10.3	-2.0	9.1	-3.2	2.2	140	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	27.7	17.4	25.8	16.7	24.1	15.8	18.8	25.4	17.7	24.2	16.6	22.5	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.5	21.9	15.1	11.7	20.4	14.2	11.0	19.3	56.2	25.4	52.5	24.3	49.2	22.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.0	7.0		-37.9	30.9	4.0	2.0	-40.8	32.3	-43.1	33.5	-45.4	34.6	-48.3	36.0
				-38.4	21.3	4.1	1.1	-41.3	22.1	-43.7	22.8	-46.0	23.4	-48.9	24.2
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-12.2	-11.9	-4.9	3.2	10.1	13.7	16.3	14.9	9.9	3.3	-5.9	-12.4		
	DBStd	12.13	8.86	7.73	7.07	5.23	4.48	3.09	3.03	3.47	4.43	4.86	6.90	7.88		
	HDD10.0	3474	687	612	461	209	55	6	0	3	55	213	478	694		
	HDD18.3	5959	946	846	719	454	257	142	78	117	254	467	728	952		
	CDD10.0	587	0	0	0	6	57	116	196	154	52	5	0	0		
	CDD18.3	29	0	0	0	0	1	3	16	9	1	0	0	0		
	CDH23.3	531	0	0	0	1	40	63	233	146	48	1	0	0		
	CDH26.7	100	0	0	0	0	5	6	54	26	9	0	0	0		
	WSAvg	3.2	3.1	3.1	3.5	3.8	3.6	3.2	2.9	2.9	3.0	3.3	3.0	2.7		
	PrecAvg	405	17	12	16	24	41	70	78	52	33	18	16	17		
	PrecMax	507	39	27	34	57	86	138	155	90	74	56	41	39		
(15) Precipitation	PrecMin	300	0	0	3	3	6	22	22	21	9	5	5	0		
	PrecStd	61	11	7	9	13	19	30	27	19	18	11	9	8		
	DB	6.6	6.9	11.5	21.7	27.2	27.3	30.8	29.4	28.0	21.8	11.2	6.0			
	MCWB	2.9	2.5	5.7	10.4	13.8	16.6	19.3	18.0	16.0	11.9	6.7	2.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	4.1	4.1	8.1	18.0	24.0	25.2	28.1	26.8	24.3	16.2	7.0	3.1			
	2% MCWB	1.5	1.0	3.5	8.1	12.4	15.7	18.5	17.2	14.9	9.5	3.7	0.5			
	5% DB	2.0	1.9	6.1	14.8	21.5	23.2	25.9	24.8	21.1	13.2	4.6	0.7			
	5% MCWB	0.0	-0.5	2.3	6.6	11.3	14.4	17.7	16.7	13.1	7.5	1.8	-1.2			
	10% DB	-0.1	-0.5	4.2	12.1	19.3	21.2	23.9	22.8	18.2	10.9	2.3	-1.6			
	10% MCWB	-1.5	-2.2	1.0	5.3	10.3	13.6	16.7	15.9	12.0	6.0	0.3	-3.0			
	0.4% WB	3.2	2.9	6.0	10.6	14.6	18.2	20.9	19.9	16.9	12.3	6.6	2.7			
	0.4% MCDB	6.1	6.2	10.3	21.2	23.4	25.3	27.1	26.3	26.1	20.4	10.7	5.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	1.7	1.1	4.0	8.5	13.3	16.7	19.3	18.4	15.2	9.7	4.1	0.7			
	2% MCDB	3.8	3.7	7.9	17.0	21.8	22.7	26.1	24.9	22.8	14.8	6.6	3.0			
	5% WB	0.2	-0.4	2.5	7.0	12.2	15.5	18.2	17.2	13.9	7.8	2.0	-1.0			
	5% MCDB	1.9	1.6	5.7	14.1	19.7	21.5	24.9	23.4	20.6	12.1	4.3	0.7			
	10% WB	-1.6	-2.4	1.3	5.7	10.9	14.4	17.2	16.2	12.5	6.4	0.2	-2.7			
	10% MCDB	-0.2	-0.6	4.0	11.3	18.0	19.9	23.0	21.9	17.6	10.0	1.8	-1.4			
	MDBR	9.8	11.2	10.1	11.3	13.0	12.1	12.6	12.7	12.4	10.1	8.3	8.9			
	5% DB	10.8	13.7	11.6	16.0	16.3	15.5	16.0	16.0	18.0	14.8	10.3	10.4			
(36) Mean Daily Temperature Range	MCWBR	8.6	11.1	7.7	8.5	7.6	7.5	7.9	7.8	9.2	8.6	7.1	8.6			
	MCDBR	10.8	13.4	11.0	15.2	14.9	13.3	14.7	14.6	16.1	13.4	9.5	10.5			
	MCWBR	8.9	10.9	7.5	8.4	7.2	7.0	7.6	7.6	8.4	8.2	7.0	8.8			
	MCDBR	8.9	10.9	7.5	8.4	7.2	7.0	7.6	7.6	8.4	8.2	7.0	8.8			
(40) Clear-Sky Solar Irradiance	taub	0.250	0.268	0.301	0.327	0.339	0.352	0.352	0.361	0.307	0.278	0.248	0.236			
	taud	2.283	2.303	2.315	2.380	2.401	2.379	2.426	2.351	2.511	2.561	2.454	2.279			
	Ebn at Noon	724	837	881	903	907	895	888	855	873	831	745	665			
	Edh at Noon	56	77	97	105	109	113	106	107	80	60	47	45			
(44) All-Sky Solar Radiation	RadAvg	0.87	1.89	3.41	4.66	5.68	5.95	6.02	4.94	3.41	1.92	0.90	0.63			
	RadStd	0.10	0.14	0.26	0.39	0.46	0.40	0.32	0.35	0.35	0.25	0.10	0.05			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	N/A
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page